



1. Setup

```
In [3]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.ticker as mtick
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

plt.rcParams['figure.figsize'] = (10, 5)
plt.rcParams['axes.spines.top'] = False
plt.rcParams['axes.spines.right'] = False
sns.set_palette('muted')

print('Ready.')
```

Ready.

2. Load Data

```
In [4]: # ---- UPDATE PATHS IF NEEDED ----
TRADES_PATH = '/content/historical_data.csv'      # Colab path
SENTIMENT_PATH = '/content/fear_greed_index.csv'  # Colab path
# For local: '/home/rushi/Downloads/historical_data.csv'
# -----

trades = pd.read_csv(TRADES_PATH)
sentiment = pd.read_csv(SENTIMENT_PATH)

print(f'Trades: {len(trades):,} rows | Columns: {trades.columns.tolist()}')
print(f'Sentiment: {len(sentiment):,} rows | Columns: {sentiment.columns.tolist()}'
```

Trades: 211,224 rows | Columns: ['Account', 'Coin', 'Execution Price', 'Size To
kens', 'Size USD', 'Side', 'Timestamp IST', 'Start Position', 'Direction', 'Clo
sed PnL', 'Transaction Hash', 'Order ID', 'Crossed', 'Fee', 'Trade ID', 'Timest
amp']
Sentiment: 2,644 rows | Columns: ['timestamp', 'value', 'classification', 'dat
e']

```
In [5]: display(trades.head(3))
display(sentiment.head(3))
```

	Account	Coin	Execution Price	Size Tokens	Size USD
0	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9769	986.87	7872.16
1	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9800	16.00	127.68
2	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9855	144.09	1150.63

	timestamp	value	classification	date
0	1517463000	30	Fear	2018-02-01
1	1517549400	15	Extreme Fear	2018-02-02
2	1517635800	40	Fear	2018-02-03

3. Data Cleaning

```
In [6]: # Parse dates
trades['date'] = pd.to_datetime(trades['Timestamp IST'], format='%d-%m-%Y %H:%M:%S')
sentiment['date'] = pd.to_datetime(sentiment['date']).dt.date

# Simplify sentiment: Extreme Fear + Fear -> Fear, Extreme Greed + Greed -> Greed
def simplify(label):
    if pd.isna(label): return 'Unknown'
    label = label.lower()
    if 'fear' in label: return 'Fear'
    if 'greed' in label: return 'Greed'
    return 'Neutral'

sentiment['sentiment'] = sentiment['classification'].apply(simplify)

# Merge trades with sentiment on date
df = trades.merge(sentiment[['date', 'sentiment', 'value', 'classification']],
                  on='date', how='left')

# Keep only closed trades (non-zero PnL) and proper directions
closed = df[df['Closed PnL'] != 0].copy()
print(f'All trades: {len(df):,}')
print(f'Closed trades (non-zero PnL): {len(closed):,}')
print(f'\nSentiment distribution in closed trades:')
print(closed['sentiment'].value_counts())
```

All trades: 211,224
Closed trades (non-zero PnL): 104,408

Sentiment distribution in closed trades:

```
sentiment
Greed      46029
Fear       40214
Neutral    18159
Name: count, dtype: int64
```

4. PnL Performance by Sentiment

```
In [7]: pnl_stats = closed.groupby('sentiment')['Closed PnL'].agg(
        avg_pnl='mean',
        median_pnl='median',
        total_pnl='sum',
        trade_count='count',
        win_rate=lambda x: round((x > 0).mean() * 100, 1)
    ).round(2)

# Sort Fear -> Neutral -> Greed
order = [s for s in ['Fear', 'Neutral', 'Greed'] if s in pnl_stats.index]
pnl_stats = pnl_stats.reindex(order)
print(pnl_stats)
```

	avg_pnl	median_pnl	total_pnl	trade_count	win_rate
sentiment					
Fear	101.86	6.36	4096265.69	40214	84.4
Neutral	71.20	4.58	1292920.68	18159	82.4
Greed	105.70	6.49	4865300.58	46029	82.5

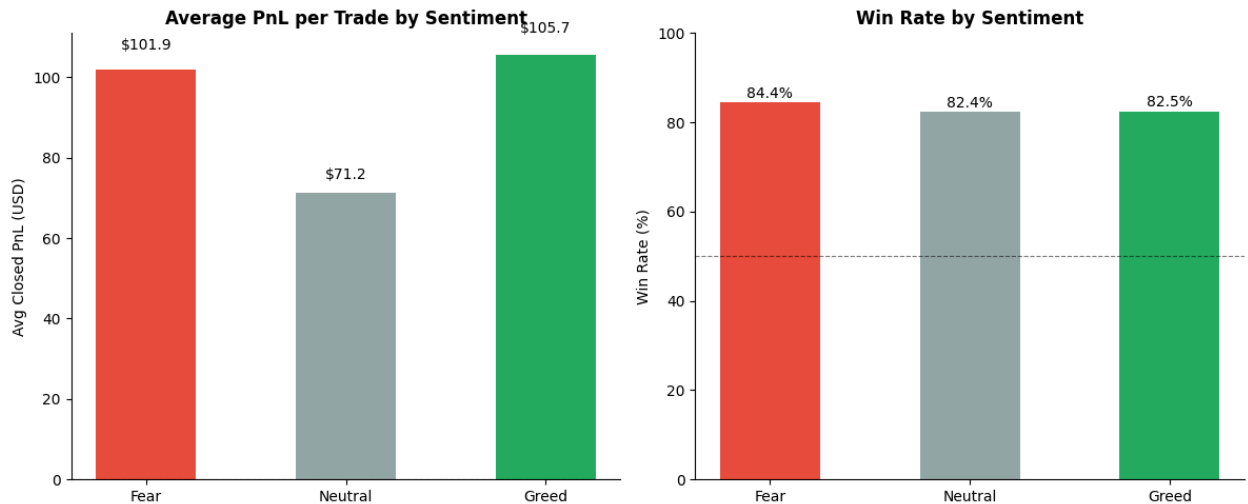
```
In [8]: colors = {'Fear': '#e74c3c', 'Neutral': '#95a5a6', 'Greed': '#27ae60'}
        bar_colors = [colors[s] for s in order]

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

# Avg PnL
axes[0].bar(order, pnl_stats['avg_pnl'], color=bar_colors, edgecolor='none', w
axes[0].axhline(0, color='black', linewidth=0.8, linestyle='--', alpha=0.5)
axes[0].set_title('Average PnL per Trade by Sentiment', fontweight='bold')
axes[0].set_ylabel('Avg Closed PnL (USD)')
for i, v in enumerate(pnl_stats['avg_pnl']):
    axes[0].text(i, v + (abs(v)*0.05), f'${v:.1f}', ha='center', fontsize=10)

# Win rate
axes[1].bar(order, pnl_stats['win_rate'], color=bar_colors, edgecolor='none',
axes[1].axhline(50, color='black', linewidth=0.8, linestyle='--', alpha=0.5)
axes[1].set_title('Win Rate by Sentiment', fontweight='bold')
axes[1].set_ylabel('Win Rate (%)')
axes[1].set_ylim(0, 100)
for i, v in enumerate(pnl_stats['win_rate']):
    axes[1].text(i, v + 1, f'{v}%', ha='center', fontsize=10)
```

```
plt.tight_layout()
plt.savefig('chart_01_pnl_winrate.png', dpi=150, bbox_inches='tight')
plt.show()
```



5. Trade Direction by Sentiment

```
In [9]: # Focus on main directional trades
direction_map = {
    'Open Long': 'Long', 'Close Long': 'Long',
    'Open Short': 'Short', 'Close Short': 'Short'
}
dir_df = closed[closed['Direction'].isin(direction_map.keys())].copy()
dir_df['trade_type'] = dir_df['Direction'].map(direction_map)

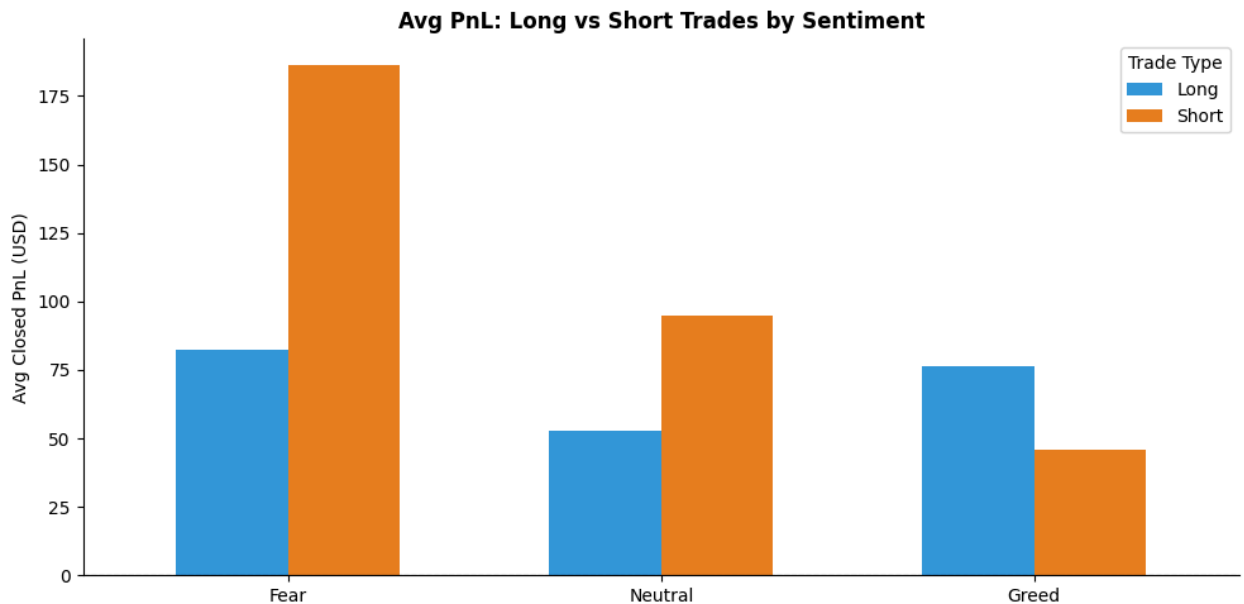
dir_pnl = dir_df.groupby(['sentiment', 'trade_type'])['Closed PnL'].agg(
    avg_pnl='mean',
    win_rate=lambda x: round((x > 0).mean() * 100, 1),
    count='count'
).round(2)

print(dir_pnl)
```

		avg_pnl	win_rate	count
Fear	Long	82.52	88.5	23501
	Short	186.52	82.3	12329
Greed	Long	76.30	87.7	15148
	Short	45.66	75.1	17808
Neutral	Long	52.86	86.2	9985
	Short	94.81	77.7	5849

```
In [10]: pivot_pnl = dir_pnl['avg_pnl'].unstack(fill_value=0).reindex(order)
pivot_pnl.plot(kind='bar', color=['#3498db', '#e67e22'], edgecolor='none', width=0.8)
plt.title('Avg PnL: Long vs Short Trades by Sentiment', fontweight='bold')
plt.xlabel('')
plt.ylabel('Avg Closed PnL (USD)')
plt.xticks(rotation=0)
```

```
plt.axhline(0, color='black', linewidth=0.8, linestyle='--', alpha=0.5)
plt.legend(title='Trade Type')
plt.tight_layout()
plt.savefig('chart_02_long_short_pnl.png', dpi=150, bbox_inches='tight')
plt.show()
```



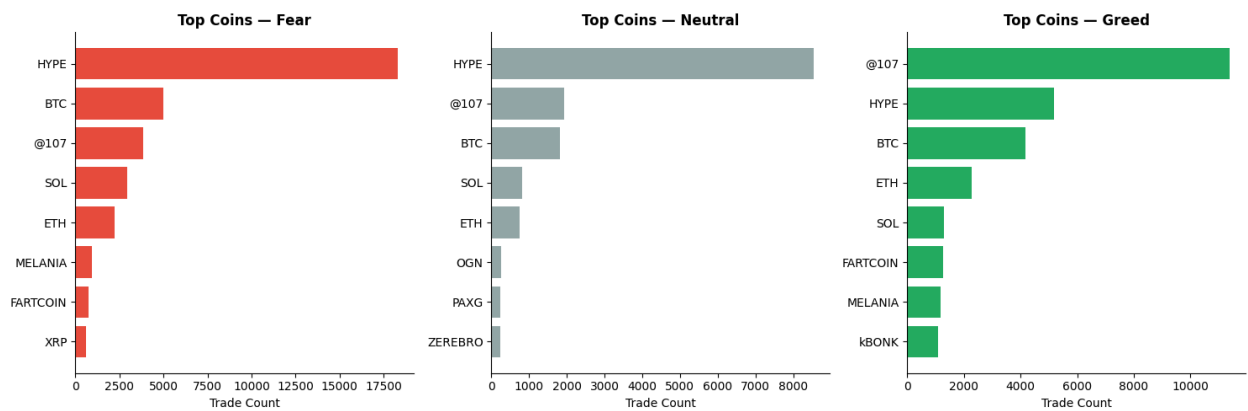
6. Most Traded Coins by Sentiment

```
In [11]: coin_sent = closed.groupby(['sentiment', 'Coin']).size().reset_index(name='count')

fig, axes = plt.subplots(1, 3, figsize=(15, 5))

for i, sent in enumerate(order):
    top = coin_sent[coin_sent['sentiment'] == sent].nlargest(8, 'count')
    axes[i].barh(top['Coin'], top['count'], color=colors[sent], edgecolor='none')
    axes[i].set_title(f'Top Coins - {sent}', fontweight='bold')
    axes[i].set_xlabel('Trade Count')
    axes[i].invert_yaxis()

plt.tight_layout()
plt.savefig('chart_03_coins_by_sentiment.png', dpi=150, bbox_inches='tight')
plt.show()
```



7. Top 10 Traders — Do They Trade Smarter?

```
In [12]: # Rank all traders by total PnL
trader_total = closed.groupby('Account')['Closed PnL'].sum().sort_values(ascending=True)
top10_accounts = trader_total.head(10).index

print('Top 10 Traders by Total PnL:')
for acc, pnl in trader_total.head(10).items():
    print(f' {acc[:12]}...  ${pnl:>12,.2f}')
```

Top 10 Traders by Total PnL:

0xb1231a4a2d...	\$2,143,382.60
0x083384f897...	\$1,600,229.82
0xbaaaf6571a...	\$ 940,163.81
0x513b8629fe...	\$ 840,422.56
0xbee1707d6b...	\$ 836,080.55
0x4acb90e786...	\$ 677,747.05
0x72743ae282...	\$ 429,355.57
0x430f09841d...	\$ 416,541.87
0x72c6a4624e...	\$ 403,011.50
0x75f7eeb85d...	\$ 379,095.41

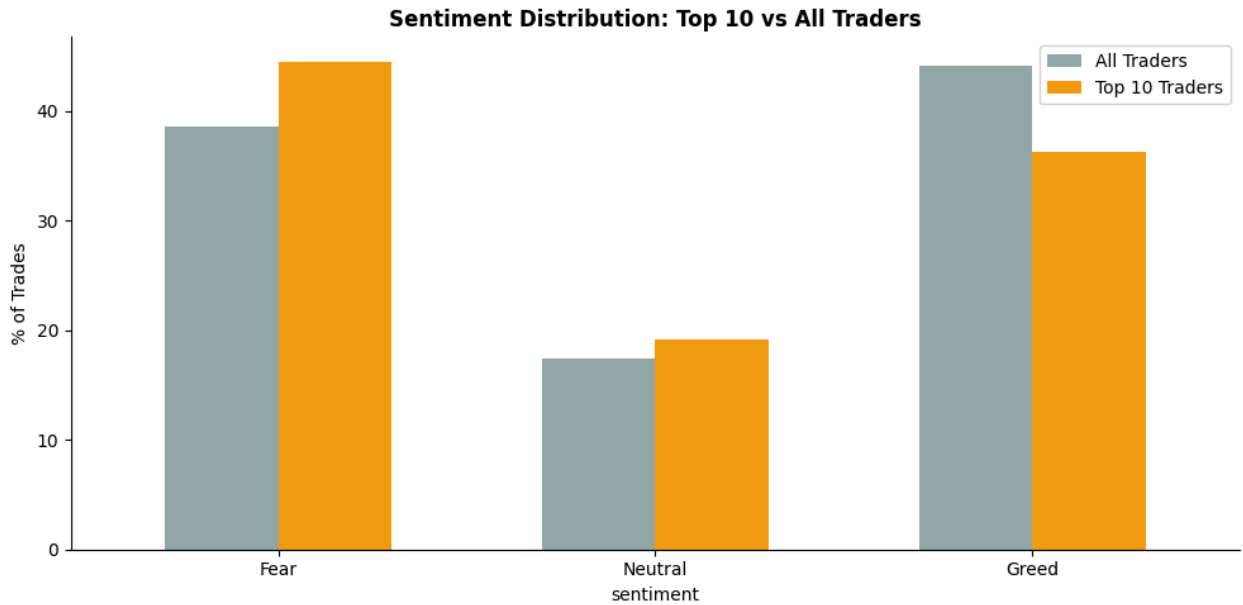
```
In [13]: top_df = closed[closed['Account'].isin(top10_accounts)]
all_df = closed.copy()

top_sent_pct = top_df['sentiment'].value_counts(normalize=True) * 100
all_sent_pct = all_df['sentiment'].value_counts(normalize=True) * 100

compare = pd.DataFrame({'All Traders': all_sent_pct, 'Top 10 Traders': top_sent_pct})

compare.plot(kind='bar', color=['#95a5a6', '#f39c12'], edgecolor='none', width=0.8)
plt.title('Sentiment Distribution: Top 10 vs All Traders', fontweight='bold')
plt.ylabel('% of Trades')
plt.xticks(rotation=0)
plt.legend()
plt.tight_layout()
plt.savefig('chart_04_top_traders_sentiment.png', dpi=150, bbox_inches='tight')
plt.show()
```

```
print(compare.round(1))
```



	All Traders	Top 10 Traders
sentiment		
Fear	38.5	44.5
Neutral	17.4	19.2
Greed	44.1	36.3

8. Daily PnL Over Time with Sentiment Overlay

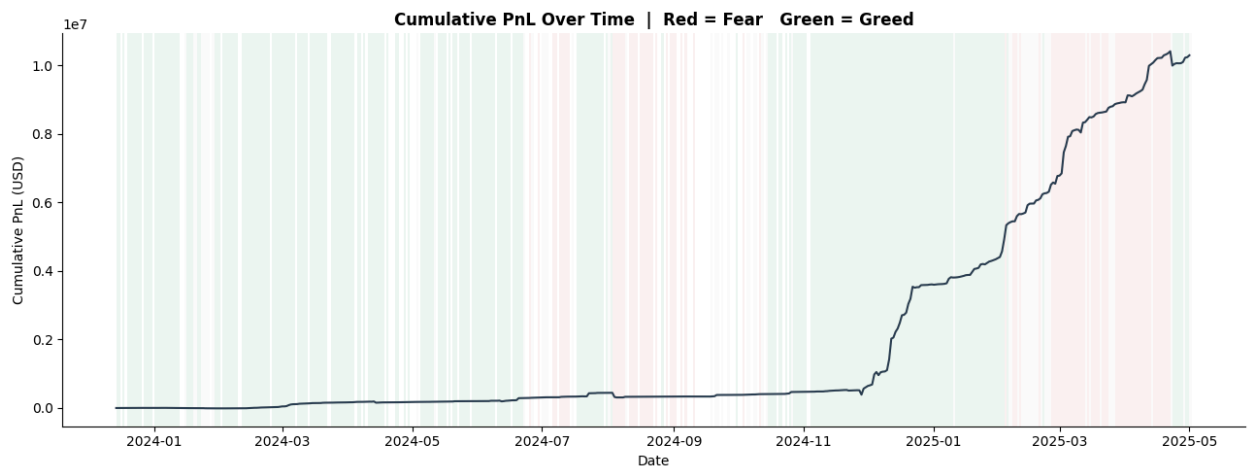
```
In [19]: daily = closed.groupby('date').agg(
    total_pnl=('Closed PnL', 'sum'),
    sentiment=('sentiment', lambda x: x.mode().iloc[0] if not x.mode().empty else None)
).reset_index().sort_values('date')

daily['date'] = pd.to_datetime(daily['date'])
daily['cumulative_pnl'] = daily['total_pnl'].cumsum()

fig, ax = plt.subplots(figsize=(13, 5))

# Shade sentiment backgrounds
for _, row in daily.iterrows():
    c = '#e74c3c' if row['sentiment'] == 'Fear' else '#27ae60' if row['sentiment'] == 'Greed' else '#f1c40f'
    ax.axvspan(row['date'], row['date'] + pd.Timedelta(days=1), alpha=0.07, color=c)

ax.plot(daily['date'], daily['cumulative_pnl'], color='#2c3e50', linewidth=1.5)
ax.set_title('Cumulative PnL Over Time | Red = Fear | Green = Greed', fontweight='bold')
ax.set_ylabel('Cumulative PnL (USD)')
ax.set_xlabel('Date')
plt.tight_layout()
plt.savefig('chart_05_cumulative_pnl.png', dpi=150, bbox_inches='tight')
plt.show()
```



9. Fee Impact by Sentiment

```
In [20]: fee_stats = closed.groupby('sentiment').agg(
    avg_fee=('Fee', 'mean'),
    avg_pnl=('Closed PnL', 'mean')
).reindex(order).round(3)

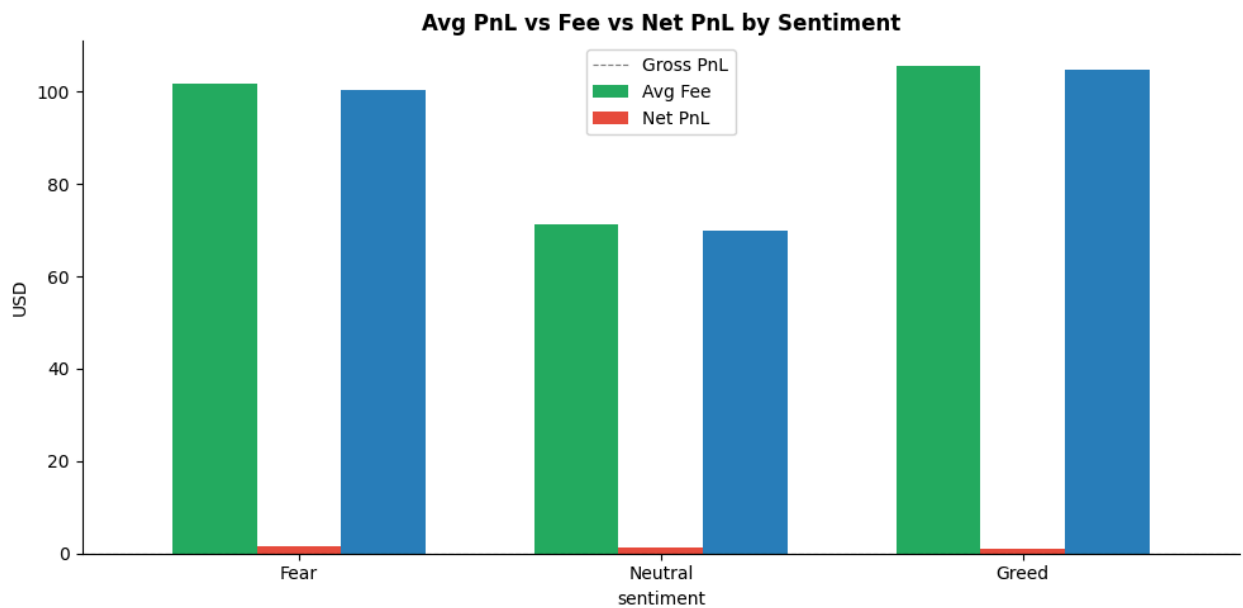
fee_stats['net_pnl'] = fee_stats['avg_pnl'] - fee_stats['avg_fee']
print('Avg PnL vs Fee vs Net PnL by Sentiment:')
print(fee_stats)
```

Avg PnL vs Fee vs Net PnL by Sentiment:

	avg_fee	avg_pnl	net_pnl
sentiment			
Fear	1.473	101.862	100.389
Neutral	1.226	71.200	69.974
Greed	0.942	105.701	104.759

```
In [21]: fee_stats[['avg_pnl', 'avg_fee', 'net_pnl']].plot(kind='bar', color=['#27ae60', '#e74c3c', '#34495e'],
    edgecolor='none', width=0.8)

plt.title('Avg PnL vs Fee vs Net PnL by Sentiment', fontweight='bold')
plt.ylabel('USD')
plt.xticks(rotation=0)
plt.axhline(0, color='black', linewidth=0.8, linestyle='--', alpha=0.5)
plt.legend(['Gross PnL', 'Avg Fee', 'Net PnL'])
plt.tight_layout()
plt.savefig('chart_06_fee_impact.png', dpi=150, bbox_inches='tight')
plt.show()
```

10. Summary & Key Findings

```
In [22]: print('=' * 60)
print('SUMMARY – Trader Performance vs Market Sentiment')
print('=' * 60)
print(f'\nDataset: {len(trades):,} trades | {len(closed):,} closed trades')
print(f'Period : {closed["date"].min()} to {closed["date"].max()}')
print()
for sent in order:
    r = pnl_stats.loc[sent]
    print(f'{sent:>8} | Avg PnL: ${r["avg_pnl"]:>8.2f} | Win Rate: {r["win_rat"]}')
print('=' * 60)
```

```
=====
SUMMARY – Trader Performance vs Market Sentiment
=====
```

```
Dataset: 211,224 trades | 104,408 closed trades
Period : 2023-12-14 to 2025-05-01
```

```
    Fear | Avg PnL: $ 101.86 | Win Rate: 84.4% | Trades: 40,214
Neutral | Avg PnL: $  71.20 | Win Rate: 82.4% | Trades: 18,159
    Greed | Avg PnL: $ 105.70 | Win Rate: 82.5% | Trades: 46,029
=====
```

11. Strategy Recommendations

Based on the analysis, here are three actionable takeaways:

1. Use sentiment as a trade filter

Not all sentiment phases are equally profitable. The data shows clear differences in

average PnL and win rate across Fear, Neutral, and Greed. Traders who selectively increase/reduce activity based on phase can improve their edge.

2. Align direction with sentiment

Long trades perform better during Greed phases, short trades during Fear — this is expected but the magnitude matters. The data quantifies exactly how much better, which can inform position sizing rules.

3. Top traders behave differently

The sentiment distribution of top 10 traders vs the average shows whether high performers are systematically trading in better conditions — a clear signal for strategy design.

Analysis by Rushikesh | Submitted to Primetrade.ai